Arista Access Node QoS/Congestion Event Troubleshooting and Resolution Procedure

Node: MALL-ACC-21 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_3048

Description

The occurrence of ALARM_3048 on the Arista access node MALL-ACC-21, situated at the Malleswaram Node, signifies a QoS/Congestion event, potentially triggered by microbursts, interface tail drops, or QoS queue starvation. This failure domain is intricately linked to the role of the access node, which acts as the ingress point for subscriber traffic. The Arista access node, configured with a sophisticated QoS framework, employs various mechanisms such as scheduling, policing, and shaping to manage traffic flows. However, the dynamic nature of modern network traffic, characterized by abrupt changes in traffic patterns and volumes, can lead to transient congestion events. These events can overwhelm the buffering capabilities of the access node's interfaces, resulting in tail drops and QoS queue starvation. Furthermore, the Arista EOS's (Extensible Operating System) QoS implementation, while highly scalable and flexible, relies on precise configuration and calibration to ensure optimal performance under diverse traffic conditions. Therefore, a meticulous and structured approach is essential to diagnose and resolve QoS/Congestion events on Arista access nodes, ensuring that the root causes are identified and addressed, and the network's performance and reliability are restored.

Pre-requisite Checks

- Verify the physical and environmental integrity of the Malleswaram Node
- Confirm the operational status of redundant links and power supplies
- Ensure that all relevant network management and monitoring systems are accessible and functional
- Dispatch remote hands to the Malleswaram Node if necessary

Potential Root Causes

Cause	Probability
Inadequate QoS configuration	High
Insufficient interface buffering	Medium
Traffic pattern changes or unexpected spikes	Low

Diagnostic Commands & Expected Outputs

```
Command: show qos interface ethernet 1/1, Expected Output: QoS interface configuration and statistics, Description: This command isolates the QoS configuration and performance metrics for the specified interface, enabling the identification of potential QoS misconfigurations or performance anomalies.
```

```
Command: show interface ethernet 1/1 counters, Expected Output: Interface traffic counters and error statistics, Description: This command provides detailed traffic and error statistics for the specified interface, helping to pinpoint issues related to interface congestion, errors, or discards.

Command: show running-config section qos, Expected Output: QoS configuration subset, Description: This command extracts the QoS-related configuration from the running configuration, facilitating the review and analysis of QoS policies, classes, and mappings.
```

Resolution Steps

Step 1: Step: Adjust QoS configuration, Command: qos schedule-profile SCHEDULE_PROFILE_NAME, Description: Apply the necessary adjustments to the QoS configuration, ensuring that the scheduling profile is correctly defined and applied to the affected interface.

Step 2: Step: Modify interface buffering, Command: interface ethernet 1/1 bandwidth 10000, Description: Increase the interface bandwidth or adjust other interface parameters to mitigate congestion and alleviate buffering constraints.

Step 3: Step: Implement traffic management, Command: policy-map type network-qos POLICY_MAP_NAME, Description: Create and apply a policy map to manage traffic flows, prioritizing critical traffic and preventing congestion-triggered drops.

Post-Check Verification

```
Command: show qos interface ethernet 1/1, Expected Output: Updated QoS interface configuration and statistics, Description: Verify that the QoS configuration changes have been successfully applied and that interface performance has improved.

Command: show interface ethernet 1/1 counters, Expected Output: Updated interface traffic counters and error statistics, Description: Confirm that the interface traffic and error statistics reflect the expected improvements following the QoS configuration adjustments.
```

Roll-back Procedures

- Step: Revert QoS configuration, Command: no qos schedule-profile SCHEDULE_PROFILE_NAME, Description: If the applied QoS configuration changes fail to resolve the issue or introduce new problems, remove the adjustments to restore the original QoS configuration.

- Step: Restore interface buffering, Command: no interface ethernet 1/1 bandwidth 10000, Description: If the interface buffering modifications are found to be ineffective or cause unforeseen issues, revert to the original interface configuration to maintain stability.